QuantMesh x402

Decentralized AI Micropayment Signal Router on Algorand

Problem Statement: PS0404 — Atomic Multi-Agent Service Router (x402)

Team: QuantMesh Innovators | College: CSBS Dept., BV(DU)COE, Pune | Event: AlgoVerse 2026 Hackathon

Problem Statement & Real-World Motivation

1. Subscription Fatigue

- Crypto signal APIs cost $100–$500/month.
- Traders pay high monthly fees even if they trade only 2–3 times.
- High financial barrier for small automated bots & retail traders.

2. Single-AI Risk

- Single AI models hallucinate or misread market volatility.
- Single-indicator tools cause heavy trading drawdown losses.
- Lack of multi-agent consensus validation.

3. Payment Friction

- Querying 4 separate AI APIs requires 4 transactions.
- High gas fees & friction on legacy chains.
- No zero-fee protection if an API fails.

Proposed Solution: QuantMesh x402 Architecture

⚡ Pay-Per-Use Micropayments ($0.007)

- Zero monthly subscription commitments.
- Users pay flat $0.007 USDC/ALGO per fused signal request.
- Pre-Execution Gating (Zero-Fee Guarantee): If any sub-agent fails, HTTP 502 returns & $0 is charged.

🤖 Multi-Agent Consensus Mesh

- Combines 4 specialized AI sub-agents in parallel:

1. Worker A: FinBERT Financial Sentiment NLP
2. Worker B: CoinGecko On-Chain Whale Net Flow
3. Worker C: Technical Indicators (RSI, SMA, MACD)
4. Worker D: Dynamic Weighted Consensus Fusion Engine

Paying Users & Sustainable Unit Economics

🎯 Target Customers & Paying Users

- Algorithmic Crypto Traders & Telegram Signal Bots.
- Web3 DeFi Protocols & DEX Interface Aggregators.
- Autonomous AI Agents needing verified market data.
- Pay-Per-Use model saves users 99.9% vs $300/mo subscriptions.

💰 Unit Economics & Protocol Margin

- User Payment: $0.0070 USDC per execution.
- Worker Payouts: $0.0060 total ($0.002 A, $0.002 B, $0.001 C, $0.001 D).
- Router Net Profit: $0.0010 USDC per call (14.3% Margin).
- At 1M calls/day = $1,000/day ($365k/year) in automated profit.

x402 Technical Protocol Flow (Challenge → Sign → Retry → Settle)

Step 1: Probe

Client sends POST /orchestrate without payment proof.

Step 2: Challenge (402)

Server pre-executes agents & returns HTTP 402 + payment headers.

Step 3: Sign (Lute)

Client reads 402 headers & signs 5-txn group in Lute Wallet.

Step 4: Settle & Result

Client retries with payment proof; GoPlausible verifies & settles in 1 block.

Full-Stack Architecture & Technology Stack

🌐 Frontend Layer

- Next.js 16 (App Router)
- Tailwind CSS Glassmorphism
- @txnlab/use-wallet-react
- Lute / Pera Testnet Signer

⚙️ Orchestrator Backend

- Hono.js High-Speed Gateway
- GoPlausible Facilitator Client
- Algorand Indexer Verification
- AWS EC2 Live Host

⛓️ Settlement & Agents

- Algorand Testnet (AVM)
- Testnet USDC ASA 10458941
- 4 Sub-Agent Services (FastAPI + n8n)
- SHA-256 Attestation Engine

On-Chain Transaction Proof & Live Verification

🔗 Live Algorand Testnet Settlement

- Router Wallet: 4DTSNS35EP...JDEHUHUNFR3KJGPO4
- Testnet USDC ASA ID: 10458941
- 5-Txn Atomic Payout Group executed in 1 single block.
- Lora Block Explorer Link: https://lora.algokit.io/testnet

🔒 GoPlausible Verification & Receipt

- GoPlausible Facilitator API (https://facilitator.goplausible.xyz) POST /verify & /settle verified.
- SHA-256 Attestation Digest: sha256(token:score:verdict:txId:timestamp).
- Live API Endpoint: https://api.dhanrajgupta.xyz/api/v1/health

Innovation & Competitive Differentiation

Traditional Signal Services

- ❌ High $300/mo subscription barrier.
- ❌ Pay upfront for potentially dead/failing APIs.
- ❌ Single-model bias & hallucination risks.
- ❌ Centralized black-box server execution.

QuantMesh x402 Solution

- ✅ $0.007 pay-per-use micropayments.
- ✅ Zero-Fee Guarantee: $0 charged if an agent fails.
- ✅ Multi-agent consensus weighted scoring.
- ✅ Decentralized 5-txn atomic settlement in 1 block.

Completeness Status & Working Deliverables

✅ Completed & Working Live

- End-to-end HTTP 402 Probe-Challenge-Sign-Retry flow.
- All 4 sub-agent microservices online on AWS EC2 & n8n.
- Lute Wallet integration with indexesToSign=[0].
- GoPlausible Facilitator /verify & /settle integration.
- SHA-256 Attestation Digest generation.

🚀 Planned Next Steps

- Deploy PyTeal ARC-4 Box Storage contract to Mainnet.
- Add dynamic worker registration portal for 3rd-party devs.
- Integrate Pyth / Chainlink live oracle feeds.
- Expand token support to 50+ crypto pairs.

Real-World Impact, Scalability & Mainnet Readiness

🌐 Real-World Impact & Ecosystem

- Democratizes high-frequency AI signal access for retail traders.
- Powers autonomous AI agents with machine-to-machine micropayments.
- Creates an open monetization marketplace for AI developers on Algorand.

⚡ Mainnet Readiness & Scalability

- Algorand Mainnet Switch: Single flag flip to algorand:mainnet.
- Mainnet USDC ASA 315667040 pre-configured.
- Sub-3-second block finality & <$0.001 network fees enable infinite throughput.